FEDERAL SPECIFICATION

WEDGES; METAL (MAGNESIUM)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers magnesium wedges used for felling timber and for bucking and splitting logs.

1.1.1 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are ordinarily used by the Federal Government.

1.1.2 *Illustrations.*—The illustrations shown in figures 1 through 4 are for the convenience of identification for requisitioning, purchasing, and inspection officers, and are not intended to preclude the purchase of wedges which are otherwise in accordance with the requirements of this specification.

1.2 *Classification.*—Magnesium wedges covered by this specification shall be of the following types, as specified:

Type I.—Falling
Type II.—Bucking
Type III.—Side pinch
Type IV.—Splitting

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications.—The following Federal specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

NN-B-591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601—Boxes; Wood-Cleated Plywood (for Domestic Shipment).
NN-B-621—Boxes; Wood, Nailed and Lock-Corner.
NN-B-631—Boxes; Wood, Wirebound (for Domestic Shipment).
QQ-M-151—Metals; General Specification for Inspection of.
LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636—Boxes; Fiber, Solid (for Domestic Shipment).

2.2 Other publication.—The following Military Standard, of the issue in effect on date of invitation for bids, forms a part of this specification:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.

(Copies of the Military standard referenced above, required by the contractor in connection with specific procurement functions, should be obtained from the procuring agency or as directed by the contracting officer.)

2.3 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—The wedges shall be cast magnesium alloy with a tensile strength of 34,000 p. s. i. (See 4.4.3.) The wedges shall have a smooth cast finish and shall be free of blow holes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects.

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3.2 Hardness.—Wedges shall be heat-treated, and shall have a Brinell hardness of not less than 55 nor more than 75. (See 4.4.2.)

3.3 Protective coating.—The wedges shall have a corrosive-resistant chromate or anodized coating.

3.4 Type I, falling.—The wedges shall be rectangular in section at the head and shall taper uniformly in thickness from head to point. The center section of the point may be thinned for ease of entering saw cut. The wedges shall be similar to either figure 1a or 1b, and shall conform to the requirements of table I.

TABLE I.—*Type I, falling magnesium wedge*

Length, +1 −¼	Width at point, +¼ −⅛	Thickness at point, +1/32 −1/16	Width at head, +⅛ −⅛	Thickness at head +1/16 −⅛	Weight, +2 −¼
Inches	*Inches*	*Inch*	*Inches*	*Inches*	*Ounces*
8	4	3/16	1½	1½	20
10	4	3/16	1½	1½	27
8	4	5/16	1¼	1¼	19
10	4	5/16	1¼	1¼	25
8	2¼	⅛	1¼	1¼	14¼

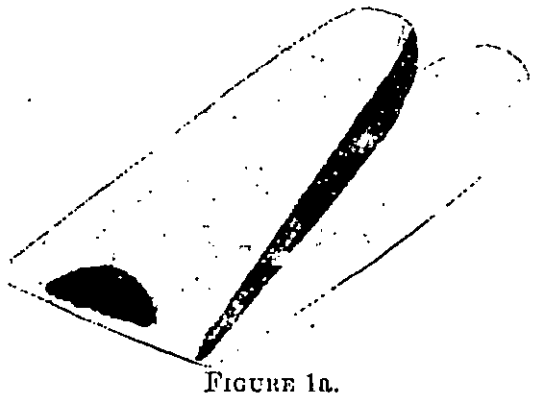


FIGURE 1a.



FIGURE 1b.—*Type I, falling magnesium wedge.*

3.5 Type II, bucking.—The wedges shall be rectangular in section at the head and shall taper uniformly in thickness from head to point. The center section of the point may be thinned for ease of entering saw cut. The wedges shall be similar to either figure 2a or 2b and shall conform to the requirements of table II.

TABLE II.—*Type II, bucking magnesium wedge.*

Length, +1 −¼	Width at point, +¼ −1/16	Thickness at point, +1/32 −1/16	Width at head, +⅛ −¼	Thickness at head, +1/8 −1/16	Weight, +2 −1
Inches	*Inches*	*Inch*	*Inches*	*Inch*	*Ounces*
8	4	3/16	1½	1	16
8	4	3/16	1½	1	16
9	3½	⅛	2¼	⅞	14

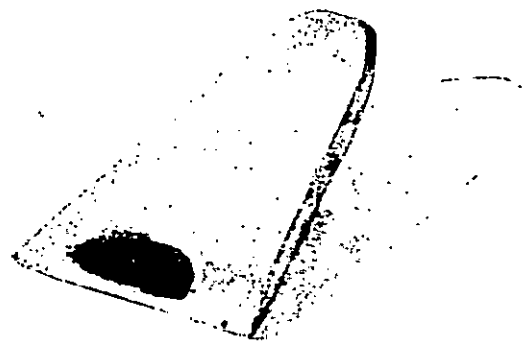


FIGURE 2a.



FIGURE 2b.—*Type II, bucking magnesium wedge.*

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3.6 **Type III, side pinch.**—The wedges shall have a ridge located in the center of each broad side, starting approximately ½ inch from point and extending approximately 5½ inches towards head. The ridge shall be approximately ¼ inch at its maximum height. The wedges shall be similar to figure 3 and shall conform to the requirements of table III.

TABLE III.—*Type III, side pinch magnesium wedge.*

Length, +1 −⅛	Width, ±⅛ −⅛	Thickness at point, +1/16 −1/32	Thickness at head +⅛ −⅛	Weight, +2 −¼
Inches 8	*Inches* 2¾	*Inch* 5/16	*Inches* 1⅛	*Ounces* 15½



FIGURE 3.—*Type III, side pinch magnesium wedge.*

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3.7 **Type IV, splitting.**—The wedges shall be rectangular in section at the head and shall taper to a point which shall not be wider than the head. The wedges shall be similar to figure 4 and shall conform to the requirements of table IV.

TABLE IV.—*Type IV, splitting magnesium wedge.*

Length, +1 −¼	Width, +½ −¼	Thickness of point, +1/32	Thickness of head, +¼ −1/16	Weight, +2 −1
Inches 8	*Inches* 2¼	*Inch* 1/16	*Inches* 2	*Ounces* 29

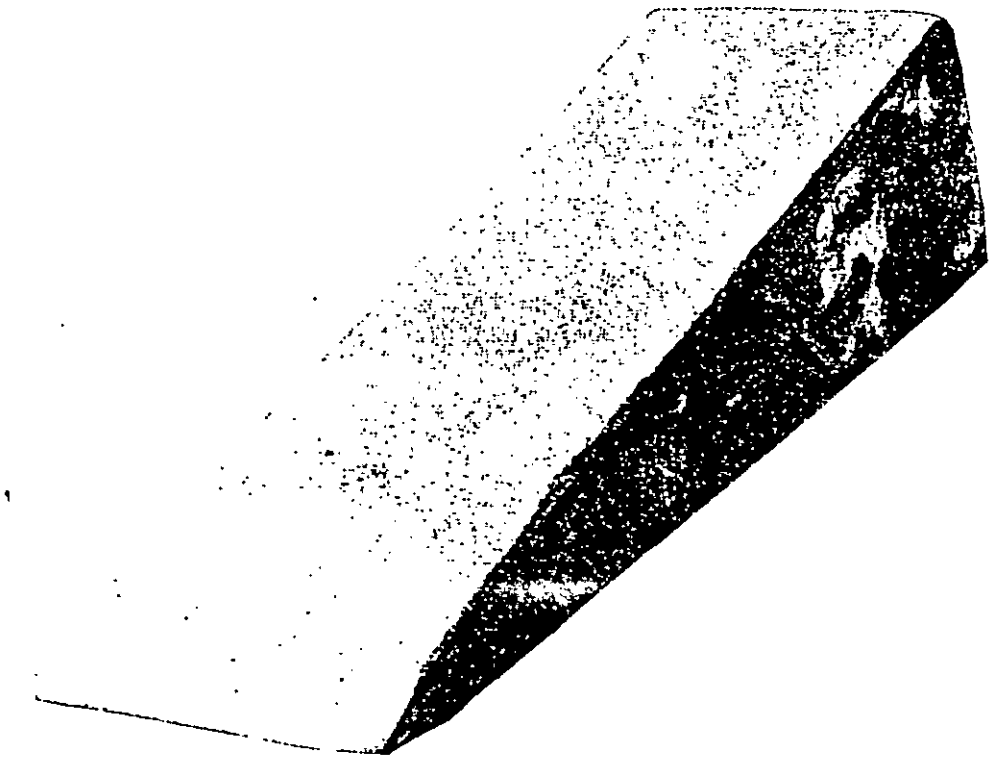


FIGURE 4.—*Type IV, splitting magnesium wedge.*

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection lot.—All wedges of the same size and type presented at one time shall be considered a lot for purposes of acceptance, inspection, and tests.

4.2 Sampling for lot acceptance.

4.2.1 *Sampling for inspection.*—A random sample of wedges shall be taken from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the following single sampling-inspection requirements in accordance with Military Standard MIL-STD-105 (see table V).

TABLE V.—*Sampling for visual and dimensional inspection, AQL (approximately)=2.5 percent defective*

Number of wedges in inspection lot	Number of wedges in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	1	2
111 to 300	25	1	2
301 to 500	35	2	3
501 to 800	50	3	4
801 to 1,300	75	4	5
1,301 and over	110	6	7

4.2.2 *Sampling for tests.*—A random sample of wedges shall be taken from each inspection lot in accordance with table VI and shall be subjected to the tests specified in 4.4.

TABLE VI.—*Sampling for lot acceptance*

Number of wedges in inspection lot	Number of wedges in the sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 110	10	1	2
111 to 300	15	1	2
301 to 500	25	2	3
501 to 800	35	3	4
801 and over	50	4	5

4.2.3 *Sampling for tension test.*—At least one sample wedge, or more, at the discretion of the Government inspector, shall be subjected to the tension test specified in 4.4.3, when applicable.

4.3 Inspection.

4.3.1 *Visual and dimensional inspection.*—Each of the sample wedges selected in accordance with table V shall be visually and dimensionally inspected by the Government inspector to verify compliance with the requirements of this specification. Any wedge in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective wedges in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedure.—Each of the sample wedges selected in accordance with table VI shall be tested as specified in 4.4.1 and 4.4.2. Any wedge in the sample which fails the tests shall be rejected. If the number of defective wedges in the sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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4.5 Rejected lots may be offered again for test provided the contractor has removed all nonconforming wedges. The Government inspector shall again take and examine samples from such resubmitted lots to verify compliance with the specification requirements. (See 4.3.1 and 4.4.)

4.4.1 *Usage test.*—A short log and a chain saw shall be provided for testing purposes. A cut shall be made in the log and the wedge inserted in the log to a depth of not more than ½ the length of the wedge. The wedge shall then be struck with an 8-pound sledge. The condition of the log shall be such that the wedge shall be struck a sufficient number of blows to provide a conclusive test. The wedge shall be rejected if it breaks, mushrooms more than ¼ inch, cracks more than 1-inch longitudinally, or if the head chips excessively.

4.4.2 *Hardness test.*—The wedge shall be tested to determine compliance with hardness requirements contained in 3.2. Wedges taken in accordance with 4.2.2 (table VI), for which the applicable hardness requirements are specified, shall be tested on a Brinell hardness tester. The reading shall be taken when using a 10 mm. ball under a 500 kg. load.

4.4.3 *Tension test.*—Each wedge taken in accordance with 4.2.3, for which the applicable physical requirements of the cast magnesium as specified in 3.1, shall be tested in accordance with the requirements of Federal Specification QQ-M-151.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—Unless otherwise specified, each package shall contain not more than 12 wedges.

5.2 **Packing.**

5.2.1 *For domestic shipment.*—Wedges processed as specified in 5.1 shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, wirebound wood, corrugated fiber or solid fiber boxes conforming to Federal Specifications NN-B-591, NN-B-601, NN-B-621, NN-B-631, LLL-B-631, and LLL-B-636, respectively.

5.3 **Marking.**

5.3.1 *Individual packages.*—Each box or package shall be plainly marked with the name of the commodity, the quantity contained therein as defined in the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the commodity, the quantity contained therein as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 **Intended use.**—This specification covers metal wedges intended for the following uses for the various types:

Type I.—Falling wedges are used in the saw kerf to fall or throw the tree to the ground.

Type II.—Bucking wedges are used after the tree has been felled and is being cut into logs. Falling wedges may be used for bucking.

Type III.—Side pinch wedges are used in the side of the log to prevent rolling.

6.2 **Ordering data.**—Purchasers should specify the type and size of wedges required and should exercise any desired options offered herein. (See 1.2, tables I through IV, 5.1, 5.3.2 and sec. 7.)

6.3 **Illustrations.**—The illustrations shown herein are included for the convenience of requisitioning and purchasing officers and are not intended to preclude the purchase of wedges which are otherwise in accordance with the requirements of this specification.

6.4 **Bid samples.**—It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with

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this specification. If, for any particular purpose, bid samples are required, invitations for bids should provide that when so requested, bidders shall submit free of expense to the Government for examination and test, samples of the magnesium wedges they propose to furnish. Samples shall remain the property of the exhibitors, but a sample may be accepted as part of a delivery in the case of a bidder awarded a contract. However, the right is reserved to destroy the sample submitted if such action is considered necessary to determine the quality of the material and method of manufacture, in which case no expense to the Government shall accrue.

6.5 Transportation description.

6.5.1 *Transportation description applicable to these items is:*

Tools, NOIBN
In barrels or boxes.
Carload weight 30,000 lbs.
Truckload weight 30,000 lbs.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Armed Forces.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN-B-591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601—Boxes; Wood-Cleated Plywood, for Domestic Shipment.
NN-B-621—Boxes; Wood, Nailed and Lock-Corner.
NN-B-631—Boxes; Wood, Wirebound (for Domestic Shipment).
LLL-B-631—Boxes; Fiber; Corrugated (for Domestic Shipment).
LLL-B-636—Boxes; Fiber, Solid (for Domestic Shipment).

Military Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
MIL-B-107—Boxes, Wood, Wirebound (Overseas Type).
JAN-P-116—Packaging and Packing for Overseas Shipment—Preservation, Methods of.

Navy Department Specification:

General Specifications for Inspection of Material.

Air Force Specification:

AN-P-13—Preservation and Packaging; Parts and Equipment (General Specification for).

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and other publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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7.2.2 *Inspection procedures (Navy only)*—General inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.3 *Preparation for delivery.*

7.2.3.1 *Cleaning, preservation, and packaging.*—Wedges shall be cleaned by method C–1 and preserved with type P–1 preservative in accordance with Military Specifications AN–P–13 and JAN–P–116. Unit packaging shall be for physical protection only. Each unit package shall contain 12 wedges.

7.2.3.2 *Packing.*—Wedges shall be packed 6 unit packages per shipping container.

7.2.3.2.1 *For domestic shipment.*—Wedges preserved and unit packaged as specified in 7.2.3.1 shall be packed in snug-fitting wood-cleated plywood, nailed wood, corrugated fiber, or solid fiber boxes, conforming to Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively.

7.2.3.2.2 *For overseas shipment.*—Wedges preserved and packaged as specified in 7.2.3.1 shall be packed in snug-fitting wood-cleated plywood, nailed wood, or wirebound wood boxes conforming to Military Specifications JAN–P–105, JAN–P–106, and MIL–B–107, respectively.

7.2.3.3 *Marking.*—Unless otherwise specified in contract or order, interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

NAVY INTEREST: *MC* Sh S

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